Stability of carotenoids, total phenolics and in vitro antioxidant capacity in the thermal processing of orange-fleshed sweet potato (Ipomoea batat...

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Intervention strategies regarding the biofortification of orange-fleshed sweet potato, which is a rich source of carotenoids for combating vitamin A deficiency, are being developed in Brazil. This study was conducted to evaluate the concentrations of individual carotenoids, total phenolic compounds and antioxidant capacity in the roots of four biofortified sweet potato cultivars that were raw or processed by four common heat treatments. HPLC, Folin-Ciocalteu, DPPH and ABTS assays were used. All cultivars showed high levels of carotenoids in raw roots, predominantly all-trans-β-carotene (79.1-128.5 mg.100 g(-1) DW), suggesting a high estimated vitamin A activity. The CNPH 1194 cultivar reported carotenoids values highest than those of other cultivars ($p < 0.05$). The total phenolic compounds varied among cultivars and heat treatments (0.96-2.05 mg.g(-1) DW). In most cases, the heat treatments resulted in a significant decrease in the carotenoids and phenolic compounds contents as well as antioxidant capacity. Processing of flour presented the greatest losses of major carotenoids and phenolics. The phenolic compounds showed more stability than carotenoids after processing. There were significant correlations between the carotenoids and phenolic compounds and the antioxidant capacity.